LANS-RISK: Queue-Aware Supervision of Learned Driving Policies under Actuation Delay

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***Abstract*—Commands selected from a current vehicle state can become inappropriate while waiting for delayed execution. This paper investigates a queue-aware, trajectory-retaining supervisor for frozen learned driving policies. LANS-RISK predicts the state reached after pending commands, incorporates a proximal policy optimization proposal into sampled future plans, and ranks plans by predicted collision-free duration, corridor retention, and tracking cost. We report the complete 44,000-record main-confirmation grid across 12 controllers, four synthetic traffic scenarios and five command delays, with focused comparisons against matched nominal MPC, PPO-only and two component removals. LANS-RISK and nominal MPC attain identical observed contiguous passing delay thresholds: 1,000 ms on Highway and Intersection, 700 ms on Merge, and 500 ms on Roundabout. Removing queue prediction reduces corridor completion in 45 of 60 paired actor/scenario/delay contrasts, all significant after the retained study-wide Holm correction. Removing the actor does not change corridor-completion rates on the evaluated grid. Tracking differences against MPC remain scenario-dependent. These findings support queue compensation within this implementation, but do not establish a general advantage of learned proposals, universal superiority over MPC, or a formal safety guarantee.**

***Index Terms*—autonomous driving, actuation delay, predictive supervision, learned control, model predictive control, reproducible evaluation**

I. Introduction

A driving controller with delayed actuation must account for commands already awaiting execution and for nearby traffic evolving before a correction arrives. Delay compensation therefore requires joint evaluation of collisions, task completion and tracking; decision time alone is insufficient.

Delay-aware control and supervision of learning-enabled controllers are established research directions. Kalaria *et al.* combine delay handling with robust tube MPC and discuss a protective role for a learning-enabled primary controller [1]. Predictive safety filtering also provides a principled framework for modifying a learned action under specified model and feasibility assumptions [2]. These precedents rule out a broad claim that predicting through delay or guarding a learned policy is itself new.

The earlier LANS manuscript [3] studies analytical path-tracking feedback with PPO steering residuals and packet-age handling. Here we evaluate trajectory supervision in interacting synthetic traffic under a common physical actuation FIFO. The contribution is an implemented design and controlled evaluation, not a new PPO algorithm or safety-filter theorem.

We study LANS-RISK, a supervisor that rolls out the pending queue, retains a shifted previous trajectory, and includes a frozen PPO proposal as a prior in sampled trajectory search. Selection prioritizes predicted collision-free duration, corridor retention and performance cost, in that order.

The conference contribution is deliberately focused: (i) an executable queue-aware trajectory-selection design with explicit information and actuation assumptions; (ii) a paired comparison with nominal MPC and PPO-only control; and (iii) matched removals of queue prediction and the actor proposal. A complete 12-controller delay table retains every main-confirmation case. Plant/traffic perturbations, stochastic observation delays and dedicated runtime evaluations remain outside this article.

II. Related Work and State-of-the-Art Positioning

A. Predictive Safety Filtering and Delayed Shielding

Learning-based MPC combines prediction and optimization with learned components [4]; safe-learning surveys distinguish empirical constraint handling from guarantees that depend on a model, uncertainty description, and safe backup [5]. Wabersich and Zeilinger [2] formulate a predictive safety filter for constrained nonlinear systems with state/input-dependent model uncertainty. Tearle *et al.* [6] specialize predictive filtering to racing: an invariant terminal set and backup trajectories support track-boundary safety, demonstrated on a 1:28-scale vehicle. Leeman *et al.* [7] use system-level synthesis to enlarge certified safe sets for linear systems with bounded disturbances.

Input latency is also addressed through sampled-data control barrier functions [8]. Cano Córdoba *et al.* [9] synthesize delay-resilient shields from safety games under worst-case observation-delay assumptions, with CARLA and gridworld demonstrations. LANS-RISK instead ranks sampled forecasts without a winning-strategy certificate or certified terminal set.

B. Delay-Aware Driving and Reinforcement Learning

Kalaria *et al.* [1] address computation/processing delay and steering dynamics with adaptive delay estimation and robust tube MPC, including a guard for a learning-enabled controller. Our fixed, known FIFO isolates delayed command execution and does not reproduce their adaptive estimator or robust tube construction.

Recent RL work emphasizes reconstructing useful information from delayed signals. Wang *et al.* [10] study delayed-observation MDPs and delay-aware training/representation strategies in continuous robotic control. Wu *et al.* [11] directly forecast beliefs for delayed RL. Random-delay RL [12] provides another formulation for delayed interactions. These methods change the learned representation or training process; the current LANS-RISK confirmation uses already frozen actors.

The supplied May 2026 version of DAROM [13] is particularly relevant to highway merging. It combines delayed observations, masked action histories, and observed delay magnitude in an encoder, and adds a stopping-distance safety controller. It evaluates stochastic V2I observation delays in SUMO using NGSIM-derived traffic parameters. Its reported success under delays up to 2 s is not directly comparable with our corridor-completion endpoint under a physical action FIFO. Earlier on-ramp studies combine RL with MPC [14] or probabilistic barrier functions [15].

C. Scope of the Comparison

Table I compares mechanisms and evidence, not numerical rankings across incompatible benchmarks. Section V lists our implemented baselines; external literature methods were not rerun. Following the motivation for unified safe-control benchmarking [16], the nominal LANS–MPC pair matches plant, structured observations, action limits and important planning resources. Other baselines have different information interfaces, as disclosed below. The benchmarks and failure metrics differ from earlier LANS [3]; here we report unpenalized episode RMSE.

III. Problem Formulation and Delay Model

A. State, Commands, and Reference Geometry

At control time t, the ego state is $x_t = (X_t, Y_t, \psi_t, v_t)$, containing planar position, heading, and speed. Surrounding-vehicle observations are $z_t = \{(X_t^i, Y_t^i, \psi_t^i, v_t^i)\}_{i=1}^{n_t}$. The model-based supervisor receives structured state and the pending command queue, but not hidden NPC desired speeds. A frozen PPO policy [17] proposes $u_t^\pi = \pi_\theta(o_t)$ from its policy observation o_t. No actor parameters change during confirmation. The legacy 18-dimensional o_t contains clipped tracking errors, a critical-NPC summary, synthetic latency/gain context and LQR advice; it omits the full NPC state and FIFO. Its synthetic timing feature is not the imposed command delay. The frozen actor is deployed directly with the inherited fusion gain forced to one and the legacy safety layer disabled. Thus PPO-only is a transferred-policy baseline with a different information representation, not an information-matched planner.

The normalized command $u_t = (\eta_t, \alpha_t) \in [-1, 1]^2$ scales to steering angle $\delta_t = 0.5\eta_t$ rad and acceleration $a_t = 3.5\alpha_t$ m/s^2. The prediction model uses wheelbase $L = 2.6$ m and step $\Delta t = 0.1$ s. For an applied command (η, α), its explicit update is

$$v^+ = \mathrm{clip}(v + 3.5\alpha\Delta t, 0, 25), \tag{1}$$

$$\psi^+ = \mathrm{wrap}(\psi + v^+ \tan(0.5\eta)\Delta t/L), \tag{2}$$

$$X^+ = X + v^+ \cos(\psi^+)\Delta t, \quad Y^+ = Y + v^+ \sin(\psi^+)\Delta t. \tag{3}$$

These equations describe the implemented ego rollout. The synthetic plant also evolves interacting NPCs, whose true future actions need not match the predictor.

Highway and Intersection use a straight reference. Merge uses a quintic lateral transition, $Y_r = -3.5 + 3.5(10\xi^3 - 15\xi^4 + 6\xi^5)$ with $\xi = \mathrm{clip}(X/25, 0, 1)$. Roundabout uses a circle of radius 8 m. Let e_y, e_ψ and κ denote the implemented lateral error, wrapped reference-heading error and reference curvature. The geometric steering prior is

$$\delta_g(x) = \arctan(L\kappa) - 0.4e_y - 1.2(e_\psi + 0.05v\kappa), \tag{4}$$

$$\eta_g(x) = \mathrm{clip}(\delta_g(x)/0.5, -1, 1). \tag{5}$$

The lateral error is $Y - Y_r(X)$ on the noncircular references and $8 - \sqrt{X^2 + Y^2}$ on the circle. Thus the method's corridor test is tied to explicit reference geometry.

B. Physical FIFO and Available Control Authority

For $d > 0$, write the commands awaiting execution as $q_t = (q_t^0, \ldots, q_t^{d-1})$, oldest first. The plant applies q_t^0 and appends the newly selected u_t:

$$x_{t+1} = F(x_t, z_t, q_t^0), \tag{6}$$

$$q_{t+1} = (q_t^1, \ldots, q_t^{d-1}, u_t). \tag{7}$$

For $d = 0$, u_t acts immediately. All methods use the same zero-filled initial FIFO, so the first d applied commands are initialization commands. The tested $d \in \{0, 2, 5, 7, 10\}$ corresponds to 0, 200, 500, 700, and 1,000 ms. A supervisory correction also enters the FIFO; it cannot replace already queued commands. This physical restriction differs from a delay-free safety override.

C. Queue Rollout and Observable Traffic Prediction

The observer estimates NPC acceleration from consecutive observed speeds:

$$\hat{a}_t^i = \mathrm{clip}((v_t^i - v_{t-1}^i)/\Delta t, -6, 1.5), \tag{8}$$

using zero on the first observation. During rollout, predicted NPC speed is clipped to $[0, 20]$ m/s, and its acceleration estimate is multiplied by 0.85 after each step. NPC heading remains fixed outside the roundabout; on the roundabout, predicted motion follows the 8 m circle. This is an observable forecast, not access to the traffic simulator's hidden intentions.

Starting from $(\hat{x}^0, \hat{z}^0) = (x_t, z_t)$, the supervisor propagates every pending command:

$$(\hat{x}^{j+1}, \hat{z}^{j+1}) = \hat{F}(\hat{x}^j, \hat{z}^j, q_t^j), \quad 0 \le j < d. \tag{9}$$

The resulting $(\bar{x}_t, \bar{z}_t) = (\hat{x}^d, \hat{z}^d)$ initializes the future-action search. At each predicted step, minimum disc clearance is $c =$

TABLE I

RELATED STATE-OF-THE-ART APPROACHES AND LANS-RISK: QUALITATIVE MECHANISM COMPARISON. LITERATURE RESULTS ARE AUTHOR-REPORTED IN DIFFERENT SETTINGS; NO EXTERNAL METHOD WAS RERUN OR RANKED AGAINST LANS-RISK. "GUARANTEE" REFERS TO THE CITED MODEL ASSUMPTIONS, NOT UNCONDITIONAL REAL-WORLD SAFETY.

Approach / source	Delay or information setting	Supervision and safety basis	Evaluation / distinction
Nonlinear PSF [2] (2021)	Model uncertainty; not our fixed FIFO protocol	Predictive constraints and safe backup under stated uncertainty assumptions	Nonlinear constrained systems; certificate differs from sampled ranking
Racing PSF [6] (2021)	State-feedback racing control	Invariant terminal set; backup trajectories; track constraints	Physical miniature vehicle; no matched traffic/FIFO experiment
Delay-aware tube MPC [1] (2022 preprint)	Estimated computation/processing delay; steering dynamics	Adaptive delay estimator and robust tube MPC	Driving/racing; richer delay model than a known queue
Delayed shielding [9] (2023)	Worst-case bounded observation delay	Safety-game synthesis and delay-resilient winning strategies	CARLA and gridworld; discrete formal abstraction
Signal-delay RL [10] (2024); DFBT [11] (2025)	Delayed observations / delayed RL information	Delay-aware representation/training or belief forecasting	Continuous robotic control; not a matched driving safety comparison
DAROM [13] (2026 version)	Stochastic V2I observation delay	Learned encoder plus stopping-distance intervention	SUMO/NGSIM-based merging; task success differs from corridor completion
LANS-RISK (this work)	Known physical FIFO, 0–1,000 ms	Queue rollout, frozen PPO prior, retained plans, lexicographic forecast ranking; no certificate	Four synthetic scenarios; paired main confirmation and component removals

$\min_i \|(X,Y)-(X^i,Y^i)\|_2 - 4.846$ m. The queue prefix is flagged if $c < 0.25$ m or $|e_y| > 1.75$ m. A prefix violation is a diagnostic of already committed commands, not a proof that the next action can recover safety.

Episodes last at most 160 steps (16 s). Highway, Merge, Intersection and Roundabout contain 5, 6, 1 and 4 NPCs, with reference speeds 15, 12, 8 and 9 m/s. Proposed initial ego speeds are uniform on [11,15], [11,15], [6,7.5] and [5.5,7.5] m/s, respectively. Goals are 80, 60 and 50 m of relative longitudinal progress for the first three scenarios (Merge also requires $|Y| < 1$ m), and one unwrapped lap for Roundabout. Goal attainment is nonterminal: logging continues until collision or the time limit. NPCs use bounded IDM-style interactions; the predictor only observes their acceleration. Initial scenes satisfy rejection-sampling predicates excluding specified overlaps and close-following configurations; these predicates do not certify solvability. Exact scene distributions and policy metadata are in the linked methods supplement. Roundabout is circular following, not entry/exit negotiation.

IV. QUEUE-AWARE TRAJECTORY SUPERVISION

A. Plan Representation and Candidate Generation

A candidate k is a sequence $r_k = \{(b_{k,h}, \alpha_{k,h})\}_{h=0}^{H-1}$ with $H = 30$. At each predicted state, it generates

$$u_{k,h} = (\mathrm{clip}(\eta_g(\hat{x}_{k,h}) + b_{k,h}, -1, 1), \alpha_{k,h}). \tag{10}$$

The offset $b_{k,h}$ permits geometric steering to adapt along a trajectory. The actor provides a first-action prior; it does not increase the common desired-speed reference v_{ref}.

Each of three rounds samples $K = 192$ plans around a per-step mean. Gaussian perturbations are shared within five-step blocks, but the mean has one value per step. All plan entries are clipped to $[-1, 1]$. Six explicit candidates replace the first sampled plans: the current mean, cruise, full brake, actor prior, retained previous plan, and best-so-far plan. Cruise uses $b_h = 0$ and $\alpha_h = \mathrm{clip}((v_{\mathrm{ref}} - \bar{v}_t)/3.5, -1, 1)$; brake uses $(b_h, \alpha_h) = (0, -1)$. The actor-prior plan replaces the cruise plan's first pair with $(\mathrm{clip}(u^\pi_{t,1} - \eta_g(\bar{x}_t), -1, 1), u^\pi_{t,2})$.

The first decision starts its mean at cruise and its retained candidate at brake. Later decisions use the shifted previous plan. Initial sampling standard deviations are $(0.18, 0.65)$ per step; the best 24 plans update the mean and standard deviation, with floors $(0.015, 0.08)$. Thus the search evaluates 576 candidate rollouts per decision, although explicit candidates may repeat. This is sampled optimization, not an exhaustive feasibility solver.

B. Collision-First Ranking and Performance Cost

For steps $h = 0, \ldots, H-1$, $c_{k,h}$ and $e_{k,h}$ are evaluated after propagation. Define the first forecast violations

$$h_c(k) = \min\big(\{h : c_{k,h} < 0.25\} \cup \{H\}\big), \tag{11}$$

$$h_e(k) = \min\big(\{h : |e_{k,h}| > 1.75\} \cup \{H\}\big), \tag{12}$$

$$s_k = \begin{cases} c_{k,0}, & h_c(k) = 0, \\ h_e(k), & h_c(k) > 0. \end{cases} \tag{13}$$

Selection lexicographically maximizes $(h_c(k), s_k, -J_k)$. A longer predicted collision-free prefix therefore takes precedence over corridor retention and tracking cost. When every sampled plan predicts a first-step collision-margin violation, first-step clearance replaces corridor duration as the secondary rank.

The implemented stage cost is explicitly

$$\begin{aligned} \ell_{k,h} = {} & 8e_y^2 + 2e_\psi^2 + 0.2(v - v_{\mathrm{ref}})^2 \\ & + 0.03(\eta^2 + \alpha^2) + 10\max(0, 1-c)^2 \\ & + \mathbf{1}_{m<0}(10^6 + 10^5 m^2), \end{aligned} \tag{14}$$

$$m = \min(c - 0.25, 1.75 - |e_y|), \tag{15}$$

$$J_k = \sum_{h=0}^{H-1} \ell_{k,h} + 0.5\|u_{k,0} - u_t^\pi\|_2^2. \tag{16}$$

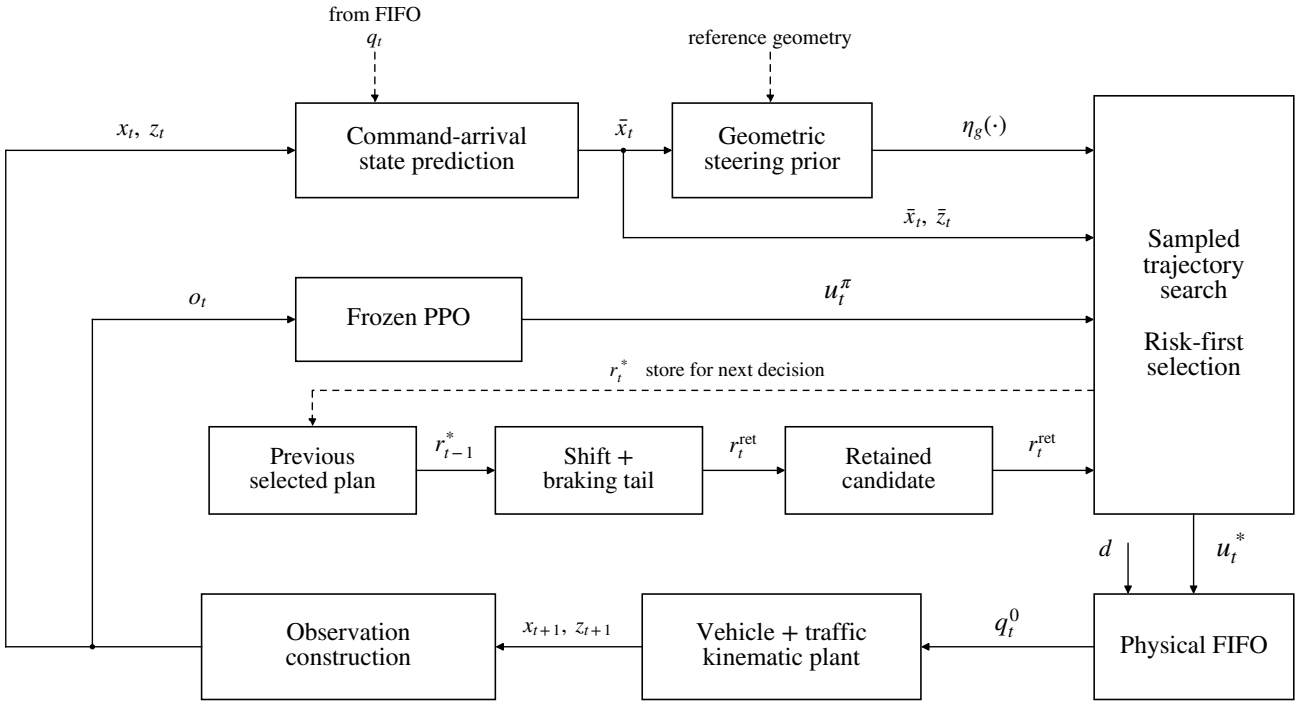


Fig. 1. LANS-RISK signal flow. Structured state (x_t, z_t) and observed NPC acceleration support queue rollout; the actor receives distinct legacy features o_t. The tall supervisor samples and ranks trajectories using arrival states, the geometric steering rule evaluated along each rollout, the PPO candidate/prior cost, and a retained plan. Dashed arrows supply queue/reference information or store the selected plan for the next decision. The first command enters the physical FIFO ($d = 0$ applies it immediately). The bottom path closes the observation loop; no EKF or LQR–PPO actuation fusion is implemented.

Variables in $\ell_{k,h}$ refer to the propagated candidate state and normalized command. These weights are frozen implementation constants, not values selected on confirmation cases. The large penalty and lexicographic rank express operational priorities; neither turns the sampled forecast into a collision-probability estimate or a safety certificate.

C. Command Execution and Plan Retention

The best plan across the three rounds supplies only its first command. For the next decision, the stored candidate is

$$r_{t+1}^{\mathrm{retained}} = (r_{t,1}^*, \ldots, r_{t,H-1}^*, (0, -1)). \tag{17}$$

The tail means geometric steering with full braking, not zero physical steering. All returned commands pass through the physical FIFO. Re-evaluation at the next observation can select a different plan. Retention therefore preserves a candidate, not a recursively feasible backup; prediction error and the uncertified braking tail prevent importing the terminal-set guarantees of [2], [6]. Algorithm 1 (Table II) summarizes the actual decision sequence.

TABLE II
ALGORITHM 1: ONE LANS-RISK CONTROL DECISION (FROZEN ACTOR).

	Inputs: observed state, FIFO q_t, PPO action u_t^{π}, retained plan.
1	Estimate NPC acceleration; roll out q_t to $(\bar{x}_t, \bar{z}_t)$; compute transient prefix diagnostics.
2	Construct cruise, brake, actor-prior and retained candidates; initialize mean and dispersion.
3	For three rounds: draw 192 plans, insert six explicit candidates, and propagate 30 steps.
4	Compute $(h_c, s, -J)$; retain the best plan across rounds. Refit the sampling distribution to 24 elites.
5	Return the selected first action to the physical FIFO. Shift the selected plan and append $(0, -1)$.
6	Update transient forecast diagnostics; repeat at the next observation.

D. Matched Component Removals

The no-prediction variant skips only the internal queue rollout; the real FIFO, actor and search remain active. The no-actor variant uses the analytical cruise proposal and sets the actor-distance weight to zero, retaining prediction, ranking and plan retention. PPO-only removes the complete supervisor and uses only the legacy policy features. Accordingly, the first two are component removals, whereas the PPO-only contrast tests a bundle. No removal establishes a claim about all prediction methods or all learned policies.

V. EXPERIMENTAL PROTOCOL

A. Frozen Confirmation and Conference Scope

The main confirmation contains 12 controllers on four scenarios and delays of 0, 200, 500, 700 and 1,000 ms. The core comparison uses LANS-RISK, nominal MPC, PPO-only, no prediction and no actor. The full delay table also includes

standalone LQR, LQR with adaptive cruise control (ACC), predictor LQR, a numerical delay-CBF filter, robust LANS, robust MPC and robust no actor. These are the project's implemented baselines, not reproductions of the cited external algorithms.

LQR variants use analytical feedback; predictor LQR adds queue prediction. Delay CBF filters the PPO proposal using numerical braking-distance constraints. The three "robust" variants sample three NPC-acceleration forecasts (nominal and ± 1 m/s^2 offsets before clipping), with 96 candidates per round. Robust MPC removes the actor and exact retained-trajectory handover; robust no actor preserves that handover. The label "robust" denotes this implemented scenario construction, not a formal robust-control guarantee. These variants do not share the nominal pair's full sampling budget.

Policies are loaded with Stable-Baselines3 [18]. Five actor-dependent variants use frozen seeds 42/43/44, and seven actor-free variants run once per condition. Each uses the same 100 traffic seeds, 141000–141099, yielding $4 \times 5 \times (5 \times 3 + 7) \times 100 = 44{,}000$ records in 440 conditions. Repeated actors reuse cases; the records are not independent pooled replicates. Development seeds are disjoint. No confirmation retraining or tuning occurred. Artifacts and the methods supplement are available at https://github.com/rizwanalihasalivi/LANS-RISK/tree/integration/aris-frozen-workflow/conference_downloads/v4. This article uses only the original main protocol, not the separate dense-Merge robustness amendment. The full-table presentation and focused contrasts were selected after evaluation and are not a separately preregistered analysis.

B. Nominal MPC Comparison

The nominal MPC shares the plant, structured state information, physical FIFO, action bounds, reference-speed ceiling, observed-acceleration predictor, 30-step horizon, and three rounds of 192 candidates. It uses a block-parameterized plan and its own cost ranking, warm start, and fallback. LANS-RISK differs in actor use, trajectory retention/representation, and ranking/objective details. Thus, the comparison matches important resources and information but is not a one-variable algorithmic ablation. Equal candidate counts also do not establish equal runtime; no computational superiority claim is made here.

C. Metrics and Statistical Analysis

Collision detection tests disc overlap at sampled 0.1 s states. Route completion requires reaching the goal and remaining collision-free for the episode. Corridor completion additionally requires $|e_t| \leq 1.75$ m at every recorded step. Episode tracking error is

$$\mathrm{RMSE} = \sqrt{\frac{1}{T}\sum_{t=1}^{T} e_t^2}. \tag{18}$$

This is conventional episode RMSE, not the failure-penalized metric in the supplied LANS path-tracking manuscript. Since collisions can shorten an episode, RMSE is interpreted together with collision and completion outcomes.

To avoid treating repeated runs or selected aggregate scores as conclusive RL evidence [19], actor-level inference and paired traffic cases are preserved. Binary rate uncertainty uses per-condition Wilson 95% intervals. Supported continuous-mean and paired-difference intervals use 5,000 traffic-case bootstrap draws. Binary comparisons use exact paired-discordance tests; RMSE uses two-sided Wilcoxon signed-rank tests with the usual symmetry assumption for a location interpretation. We retain Holm correction [20] over all 3,440 contrasts/outcomes in the parent main-confirmation analysis, including comparisons not discussed individually here. Correction is not recomputed on a favorable selected subset. Actor-seed results remain separate for inference. The full delay table and figure pool records only for descriptive display. Each scenario contributes equally; actor-free denominators are 400 per delay and actor-dependent denominators are 1,200. Inference remains within actor/scenario conditions.

The empirical delay threshold is the largest tested delay for which collision rate is at most 1% and corridor completion is at least 90% at that delay and every lower grid point. It is a finite-grid endpoint, not a continuous-delay stability bound. A passing value of 1,000 ms is right-censored by the tested grid. Even zero observed collisions in 100 cases has a Wilson 95% upper bound near 3.7%, so passing does not establish a population collision probability below 1%.

VI. Results

A. LANS-RISK versus Nominal MPC

All three LANS-RISK actors and nominal MPC attain the same observed contiguous passing thresholds: Highway 1,000 ms, Merge 700 ms, Intersection 1,000 ms, and Roundabout 500 ms. No delay-threshold advantage is established. Table III reports every tested delay, and Fig. 2 shows all controller profiles. Identical thresholds should not be read as statistical equivalence or identical episode trajectories.

At 1,000 ms, both methods complete the corridor in all Highway cases, while their actor-mean corridor rates are 35% versus 34% on Merge, 99% versus 99% on Intersection, and zero for both on Roundabout (scenario-specific companion tables). LANS-RISK has lower mean RMSE on Merge and Roundabout, while MPC has lower mean RMSE on Highway and Intersection. These differences do not establish a universal tracking winner. For Merge at 1,000 ms, actor 42 has 35/100 corridor completions (Wilson 95% interval 26.4–44.7%); MPC has 34/100 (25.5–43.7%). These marginal intervals do not test equivalence.

The full-grid table also exposes weak baselines: LQR has only 60/400 corridor completions at zero delay despite a mean RMSE of 0.039 m. Early collision termination can produce deceptively small tracking errors; Fig. 2 therefore retains collisions alongside completion.

Across the 60 per-actor/scenario/delay LANS–MPC corridor-completion contrasts, nine have a positive LANS-minus-MPC mean difference, none has a negative mean

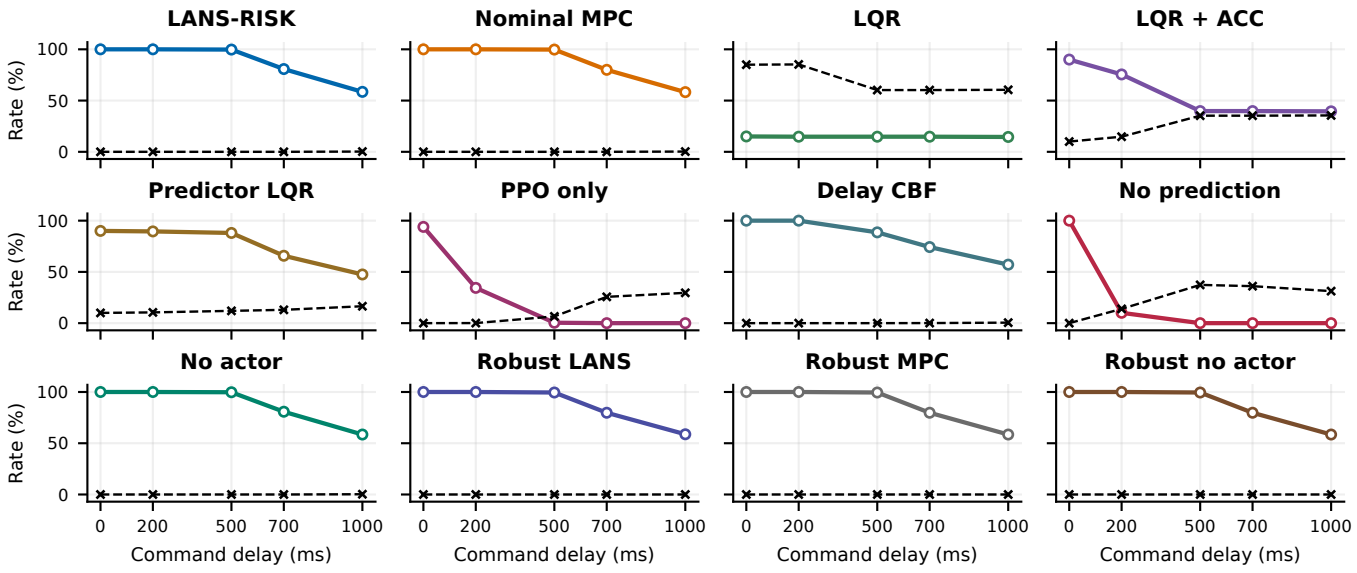


Fig. 2. All 12 controllers across the five tested command delays. Each panel isolates one controller; colored solid curves show corridor completion and black dashed curves show collisions. Rates are descriptive record averages over four scenarios and available actors, matching Table III. No outcomes are shifted. Scenario-specific failures and uncertainty remain in the companion tables and CSVs.

difference, and none is significant after the retained study-wide Holm correction. Absence of significance is not a non-inferiority or equivalence result. We did not specify an equivalence margin or conduct a dedicated equivalence design.

B. Contribution of Queue Prediction

Disabling internal queue prediction reduces corridor-completion rates in 45 of the 60 paired actor/scenario/delay contrasts; all 45 remain significant after Holm correction. No contrast has a higher corridor-completion rate for the no-prediction variant. These counts are correlated comparisons over shared cases and actors, not 60 independent replications.

The endpoint behavior illustrates the mechanism's limitations and value. At 1,000 ms, removing prediction yields zero corridor completion in every scenario. Mean collision rates rise to 25% on Intersection and 100% on Roundabout, while Highway and Merge can have zero collisions yet still fail corridor completion. This combination explains why collision-only reporting would be misleading. The physical queue remains active in the ablation; the degradation concerns planning from the wrong effective initial state within this implementation.

C. Contribution of the Actor and Supervision Bundle

Removing the actor leaves corridor-completion rates unchanged in all 60 per-actor/scenario/delay comparisons. Scenario-specific results show mixed tracking effects at 1,000 ms: the no-actor variant has slightly lower mean RMSE on Intersection and Roundabout, but higher mean RMSE on Highway and Merge. Therefore, the present evidence does not establish a general completion benefit from the learned prior.

Against PPO-only, LANS-RISK has higher corridor-completion rates in 49 of 60 contrasts, with 45 significant after Holm correction. The improvement applies to the full supervision bundle, including its richer state/FIFO interface; it does not establish a benefit over a policy trained for the current information and actuation setting. PPO-only has zero corridor completion at the largest tested delay in every scenario and high collision rates on some scenarios. This comparison cannot isolate individual supervisory effects.

VII. Discussion and Limitations

Queue-aware planning matters within this implementation, whereas the learned proposal has no observed completion benefit and nominal MPC matches its delay thresholds. Actor usefulness therefore requires measurement rather than assumption.

The cold FIFO limits control authority: initial queued commands can cause corridor departure before a correction arrives. Search cannot replace committed commands. Thresholds consequently depend on initialization and scenario geometry as well as the controller.

Plan retention and lexicographic ranking lack isolated nominal ablations, so their individual benefits are not established. The robust variants change several resources and cannot supply that missing attribution. Several boundaries limit generalization. The benchmark uses synthetic disc vehicles, a finite horizon, structured state access, fixed actors, and a known command queue. Corridor membership tests the reference point at sampled ticks, not the vehicle footprint or continuous motion. The study does not establish real-road safety, robustness to arbitrary uncertainty, or performance with perception

TABLE III

TRACKING PERFORMANCE AND CORRIDOR COMPLETION OVER ALL TESTED COMMAND DELAYS (0–1,000 MS). EACH CELL IS MEAN EPISODE RMSE (M); CORRIDOR-COMPLETED/TOTAL RECORDS, AGGREGATED OVER FOUR SCENARIOS. CORRIDOR COMPLETION REQUIRES ROUTE COMPLETION, NO COLLISION AND NO CORRIDOR EXCURSION. ACTOR-DEPENDENT COLUMNS CONTAIN 1,200 RECORDS PER DELAY; ACTOR-FREE COLUMNS CONTAIN 400. ACTOR REPETITIONS SHARE TRAFFIC CASES. NO INTERMEDIATE DELAY WAS MEASURED.

Delay (ms)	LANS-RISK	Nominal MPC	LQR	LQR + ACC	Predictor LQR	PPO only
0	0.063; 1200/1200	0.028; 400/400	0.039; 60/400	0.019; 360/400	0.019; 360/400	0.402; 1127/1200
200	0.066; 1200/1200	0.031; 400/400	0.448; 59/400	0.234; 302/400	0.019; 358/400	1.787; 413/1200
500	0.115; 1197/1200	0.092; 399/400	1.726; 59/400	2.080; 159/400	0.093; 352/400	8.739; 5/1200
700	0.192; 969/1200	0.174; 320/400	2.261; 59/400	2.369; 159/400	0.202; 263/400	12.440; 0/1200
1000	0.357; 702/1200	0.353; 233/400	5.232; 58/400	5.102; 158/400	0.429; 190/400	16.136; 0/1200

Delay (ms)	Delay CBF	No prediction	No actor	Robust LANS	Robust MPC	Robust no actor
0	0.398; 1200/1200	0.063; 1200/1200	0.061; 400/400	0.074; 1200/1200	0.044; 400/400	0.070; 400/400
200	0.650; 1200/1200	3.586; 121/1200	0.064; 400/400	0.076; 1200/1200	0.046; 400/400	0.072; 400/400
500	1.000; 1063/1200	17.263; 0/1200	0.112; 399/400	0.126; 1194/1200	0.102; 398/400	0.123; 398/400
700	1.103; 891/1200	23.461; 0/1200	0.188; 323/400	0.207; 957/1200	0.189; 319/400	0.206; 319/400
1000	1.423; 685/1200	11.365; 0/1200	0.358; 234/400	0.405; 705/1200	0.384; 234/400	0.402; 234/400

failures. Sampling only finitely many trajectories cannot certify feasibility; retaining a plan does not create a safe terminal set. Information about queue-prefix violations is diagnostic and should not be advertised as a guarantee of recovery.

The analysis preserves failures, paired cases, and frozen source/model hashes. Saved logs contain episode metrics and flags, not full state/action trajectories. Aggregate statistics and tests are recomputable, but individual RMSE, collision/corridor events and queue histories cannot be reconstructed independently from these logs. Transient forecast diagnostics are not all persisted. Confidence intervals are marginal, actor repetitions share traffic cases, and the fixed set of three actor seeds does not characterize all possible training outcomes. Shared-session episode timings are not evidence of real-time execution guarantees.

VIII. CONCLUSION

We presented and evaluated a queue-aware, trajectory-retaining supervisor for frozen learned driving policies under delayed command execution. The 44,000-record main grid and focused matched comparisons support the importance of queue prediction within this design, while revealing no corridor-completion benefit from the actor prior and no observed delay-threshold advantage over matched nominal MPC. Tracking differences are scenario-dependent, and failures remain visible. The resulting contribution is an empirically characterized supervision design and a reproducible comparison, with explicit limits on novelty, safety, and generalization.

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